Study Report: Practical Malware Analysis - Cybersecurity Analyst

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Family:	Cybersecurity Analyst
Level:	Advanced
Chapters:	7
Source:	Reference: Practical Malware Analysis - Michael Sikorski & Andrew Honig

Official sources and free libraries

- https://nostarch.com/malware

Summary

Study report for *Practical Malware Analysis* by Michael Sikorski & Andrew Honig (Advanced level) mapped to the Cybersecurity Analyst role. Reverse engineering and malware triage for security analysts.

Chapters

Track: Book-Intro

Chapter 1: About Practical Malware Analysis [Advanced]

Chapter focus: About Practical Malware Analysis *(Level: Advanced)**

This study report summarizes *Practical Malware Analysis* by Michael Sikorski & Andrew Honig for the Cybersecurity Analyst role. The resource is rated Advanced level. Reverse engineering and malware triage for security analysts. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Practical Malware Analysis
# Author: Michael Sikorski & Andrew Honig
# Role: Cybersecurity Analyst
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on Practical Malware Analysis.

When to use it
When learning Cybersecurity Analyst skills at Advanced level.

Where to use it
Cybersecurity Analyst training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role *(Level: Advanced)**

Cybersecurity Analyst professionals use ideas from Practical Malware Analysis to solve real workplace problems. Reverse engineering and malware triage for security analysts. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Cybersecurity Analyst
Book focus: Reverse engineering and malware triage for security analysts.
Recommended level: Advanced
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Cybersecurity Analyst bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered *(Level: Advanced)**

The main topics in Practical Malware Analysis include practical concepts described as: Reverse engineering and malware triage for security analysts. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Cybersecurity Analyst jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Zero Trust Architecture [Advanced]

Chapter focus: Zero Trust Architecture *(Level: Advanced)**

Zero trust assumes breach: verify explicitly, least privilege, assume breach. Identity becomes perimeter via Conditional Access, device compliance, and microsegmentation. Replace flat network VPN with app-level access policies.

Code Reference:

```
policy = {'user': 'teacher', 'device': 'compliant', 'app': 'gradebook', 'action': 'allow'}
```

What it shows: Policy tuples express zero trust decision logic.

Topic guide

What it actually is
Zero trust modernizes perimeter-centric security models.

When to use it
Cloud migration, remote work, and high-value data protection.

Where to use it
Azure AD Conditional Access, Zscaler, BeyondCorp patterns.

Real use example
Non-compliant phone blocked from admin portal even on corporate Wi-Fi.

Chapter 5: Applied: Threat Hunting with Python [Advanced]

Chapter focus: Threat Hunting with Python *(Level: Advanced)**

Proactive hunts query SIEM and endpoint data for hypotheses: uncommon parent-child pairs, rare process command lines, beaconing intervals. Python pandas ingests exported logs for statistical anomaly detection beyond static rules.

Code Reference:

```
import pandas as pd
ev = pd.read_json('edr_events.jsonl', lines=True)
rare = ev.groupby('process').size().sort_values().head(20)
```

What it shows: pandas surfaces rare processes worthy of manual investigation.

Topic guide

What it actually is
Threat hunting finds undetected adversary activity.

When to use it
Mature SOCs after baseline detections are tuned.

Where to use it
Fortune 500 SOCs and critical infrastructure CSIRTs.

Real use example
Hunt finds periodic 90-second HTTPS beacons missed by threshold alerts.

Chapter 6: Applied: Cloud Security and IAM [Advanced]

Chapter focus: Cloud Security and IAM *(Level: Advanced)**

Misconfigured S3 buckets, overly broad IAM roles, and public security groups cause breaches. Enforce least privilege, MFA on root, CloudTrail logging, and CSPM scanning. Understand shared responsibility model per provider.

Code Reference:

```
# aws iam simulate-principal-policy (conceptual)
# Azure PIM for just-in-time admin
```

What it shows: IAM simulation tests effective permissions before deployment.

Topic guide

What it actually is
Cloud security secures ephemeral infrastructure and identities.

When to use it
AWS/Azure/GCP migrations and multi-cloud estates.

Where to use it
SaaS backends and data lakes storing learner analytics.

Real use example
CSPM flags public S3 bucket with grade exports-emergency lockdown.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice *(Level: Advanced)**

Finish Practical Malware Analysis with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Cybersecurity Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it
CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.